What Ten Industrial Surveys Identify in a Cone-Crusher Population Balance Model: Bayesian Calibration with Structural Discrepancy

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***Abstract*. A cone-crusher model that predicts throughput, power and product size rests on a handful of fitted scalars, and a point fit to a few industrial surveys does not say which of them the surveys determine. We calibrate the eight fitted scalars of a Whiten classification-breakage population balance engine of a Metso HP500 cone crusher (capacity, Bond work index, power form, classification and breakage constants) as a Bayesian posterior against the ten surveys of Rocha et al., with a linear structural discrepancy on the closed-side setting and the feed size in the form of Kennedy and O'Hagan. The posterior is sampled with the differential-evolution moves of ter Braak, after the affine-invariant stretch move failed an autocorrelation-time criterion on the same posterior. Strict leave-one-survey-out validation refits every quantity without the held-out survey. Held out, the posterior predictive mean has a mean absolute percentage error of 12.00 % on throughput and 14.31 % on power, which is within the noise of a setting-only line (13.67 % and 12.78 %) and no gain over it on power. The 80 % bands cover 0.8 and 0.9 of the surveys, and on power the band is 43.9 % narrower than the residual band of the refit backbone at the same coverage; against the setting-only line (width 119.60 kW, coverage 0.8) and the free-form fit (129.82 kW, 0.8) its width of 117.74 kW is about the same, with a higher coverage, so the band gain is against the backbone and not against the simple line. Of the 16 parameters, 7 are identified by the surveys; the classification and breakage scalars of the population balance stay at their prior. With ten surveys and no measured product size, the data calibrate how the two outputs depend on the setting, and they cannot calibrate the population balance.**

***Index Terms*. Cone crusher, population balance, Bayesian calibration, model discrepancy, identifiability, leave-one-out validation, differential-evolution MCMC, comminution.**

I. Introduction

THE electricity of a copper concentrator goes especially to crushing and grinding, and in Chile in 2024 the concentrators took 55.7 % of the electricity of copper mining [1]. A cone crusher is modelled in the tradition of Whiten [2]: a classification function decides which particles are broken, a breakage function distributes the progeny, and a mass balance gives the product. The classification constants, the breakage law, the capacity and the power form of such an engine are fitted to survey data, and the fitted values are then used as if they were known.

Industrial surveys of a crusher are few and expensive. Rocha et al. [6] report ten surveys of an HP500 secondary cone crusher on itabirite iron ore with five numbers each: the closed-side setting (CSS), the feed size f_{80}, the throughput and the motor power, and a product size distribution that appears only as a figure. A point fit to ten rows of two measured outputs raises two questions that the point fit does not answer. Which of the fitted scalars do the surveys determine, and which does the fit leave where it started? And how well calibrated is the prediction of an unseen survey?

We answer both with a Bayesian calibration of the engine with an explicit structural discrepancy. The engine is run as its own simulator (a few milliseconds per evaluation), the eight fitted scalars get weakly informative priors centred on the values of Rocha et al., and a linear discrepancy on the setting and the feed size is added to each measured output in the form of Kennedy and O'Hagan [9]. The result is a posterior per scalar, a predictive distribution per held-out survey and a table of prior against posterior. Our contributions are:

1) a Bayesian calibration with discrepancy of a cone-crusher population balance engine, with the discrepancy integrated out exactly so that the sampler explores ten dimensions only;
2) a sampler configuration chosen by a measured criterion: the differential-evolution moves meet an autocorrelation-time criterion that the stretch move does not meet on the same posterior;
3) a strict leave-one-survey-out evaluation over the ten surveys, against a setting-only line, a refit backbone with a residual band and a free-form linear fit, with calibration (coverage and width) scored beside accuracy;
4) the identifiability table: which scalars the surveys determine, which they determine partly and which stay at their prior, and the structural discrepancy that the power output requires.

The calibrated prediction is not more accurate than a line in the closed-side setting. The result is the identifiability finding

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Interactive studio and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ChancaDEM.

and a calibrated predictive band, not a gain in point accuracy.

II. Related Work

A. Calibration with discrepancy

Kennedy and O'Hagan [9] calibrate the parameters of a computer model against field data by writing the data as the simulator at the true parameters plus a model discrepancy, itself a Gaussian process, plus measurement error. Higdon et al. [10] extend the construction to a simulator that cannot be run at every parameter value, with Gaussian process models of the unknown functions, and Bayarri et al. [11] give a six-step Bayesian framework for validating a computer model against reality. Brynjarsdóttir and O'Hagan [12] show that an analysis that ignores the model discrepancy can give biased and over-confident parameter estimates and predictions, and that the confounding of the discrepancy with the calibration parameters is resolved only with meaningful priors. We use the low-order form: the discrepancy is linear in the standardized setting and feed size, with a Gaussian prior on its three coefficients per output. The engine is cheap, so no emulator is used.

B. Samplers

The affine-invariant ensemble sampler of Goodman and Weare [13], implemented in `emcee` [14], moves a walker along the line to another walker (the stretch move). ter Braak [15] proposed differential-evolution Markov chain Monte Carlo, a multiple-chain sampler whose jumps are built from differences between chains, and ter Braak and Vrugt [16] added a snooker updater so that fewer than twice the dimension of chains suffice. Both families are gradient free, which matters here because the engine solves a linear system per evaluation and is written in plain Python.

C. Crusher models and their calibration

Rocha et al. [6] fitted a model of the HP500 on the same ten surveys and published the fitted expressions for the classification constants, the breakage parameter, the capacity and the power form; they are the nominal values of our priors. Lindqvist and Li [7] present a cone-crusher model with a coupled energy and breakage dependency. Morrell [8] gives a size-specific energy for crushing circuits that we use as a comparator for power. The engine itself combines the Whiten balance [2], the JKMRC breakage and classification relations [3], Evertsson's capacity model [5] and Bond's law [4]. Literature searches in Crossref and on the open web for a Bayesian calibration with discrepancy of a crusher or mill population balance model returned general methodology and no application to a cone crusher evaluated on held-out surveys; the search is not exhaustive, because Scopus and Web of Science were not searched.

III. Method

A. Data

The data are the ten surveys of the Metso HP500 secondary cone crusher at an itabirite iron-ore operation (Table V of Appendix B): the closed-side setting spans 38 to 55 mm, the feed f_{80} 47 to 140 mm, the throughput 813 to 1639 t/h and the motor power 187 to 355 kW. Throughput and power are the two targets. The product P_{80} is not measured (the product size distributions of the source are figures only), so it is never a target.

B. The engine

For a survey with setting c and feed size f_{80} the engine evaluates a classification-breakage balance on a size grid. The classification function is zero below K_1, one above K_2 and

$$C(d) = 1 - \left(\frac{K_2 - d}{K_2 - K_1} \right)^{K_3} \qquad (K_1 < d < K_2), \tag{1}$$

with $K_1 = s_{K_1}(0.23\,c + 0.30\,f_{80})$ and $K_2 = \max\{s_{K_2}(12.0 + 0.55\,c + 0.40\,f_{80}),\ 1.05\,K_1\}$ in mm, the coefficients being those of Rocha et al. [6] and s_{K_1}, s_{K_2} multipliers. The breakage function follows from the t_{10} relation $t_{10}\,(\%) = 64 + \Delta t_{10} - 0.12\,c - 0.23\,f_{80}$, capped at 44, through the appearance function of Austin. The product mass fractions are $\mathbf{p} = (\mathbf{I} - \mathbf{C})(\mathbf{I} - \mathbf{BC})^{-1}\mathbf{f}$ for a feed $\mathbf{f}$ reconstructed from the measured f_{80} with a Rosin–Rammler distribution of uniformity 0.9. The throughput depends on the setting and one scalar,

$$Q = Q_{\mathrm{ref}}\ \sigma(c), \tag{2}$$

where $\sigma(c)$ is the capacity factor of the machine, linear in the setting, at the fixed throw of 34 mm and speed of 360 rpm. The Bond power and the calibrated power are

$$P_d = 10\,W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{f_{80}}} \right) Q, \tag{3}$$

$$P_c = \zeta\,P_d + P_n, \tag{4}$$

with the sizes in micrometres, W_i the Bond work index of the ore blend, and ζ and P_n the slope and the no-load power of the current-based form of Rocha et al. [6]. The eight scalars $\boldsymbol{\theta} = (Q_{\mathrm{ref}}, W_i, \zeta, P_n, s_{K_1}, s_{K_2}, K_3, \Delta t_{10})$ are the quantities calibrated. Two facts of the structure matter for what the surveys can identify. Throughput depends on Q_{ref} alone: the other seven scalars never enter it. Power depends on all eight, the four of the population balance (s_{K_1}, s_{K_2}, K_3, Δt_{10}) only through the product size P_{80} in the Bond term, and W_i and ζ only through their product.

C. Priors

Each scalar has a truncated normal prior around its nominal value, on the logarithmic scale for the positive scalars (Table I). The nominal values are those of Rocha et al. [6] (Tables 4 and 5) and, for W_i, the blend of the compact and supercompact itabirite bands (7.2 and 10.6 kWh/t, Table 1 of the same work). The widths are weakly informative choices that bracket the spans quoted there; we found no published range of these fitted scalars for this ore, so the widths are ours. The noise scale of each output has $\ln \sigma_t \sim N(\ln m_t, 1)$ with medians $m_{\mathrm{tph}} = 60$ t/h and $m_{\mathrm{kW}} = 15$ kW, about 5 % of the typical level.

TABLE I
PRIORS OF THE EIGHT ENGINE SCALARS. THE SAMPLED SCALE IS THE NATURAL LOGARITHM FOR THE POSITIVE SCALARS.

Scalar	Unit	Nominal	Prior (sampled scale)	Range
Q_{ref}	t/h	647.00	$\ln x \sim N(6.47,\, 0.25^2)$	300.00 to 1400.0
W_i	kWh/t	9.00	$\ln x \sim N(2.20,\, 0.17^2)$	5.00 to 14.0
ζ	–	1.30	$\ln x \sim N(0.26,\, 0.23^2)$	0.50 to 2.5
P_n	kW	110.00	$N(110.0,\, 50.0^2)$	0.0 to 300.0
s_{K_1}	–	1.00	$\ln x \sim N(0.00,\, 0.15^2)$	0.50 to 2.0
s_{K_2}	–	1.00	$\ln x \sim N(0.00,\, 0.15^2)$	0.50 to 2.0
K_3	–	2.30	$N(2.3,\, 0.5^2)$	1.0 to 4.0
Δt_{10}	points of t_{10}	0.00	$N(0.0,\, 4.0^2)$	−15.0 to 15.0

D. Discrepancy and likelihood

For survey i and output $t \in \{\mathrm{tph}, \mathrm{kW}\}$,

$$y_{t,i} = f_t(\boldsymbol{\theta}; c_i, f_{80,i}) + \delta_t(\mathbf{z}_i) + \varepsilon_{t,i}, \quad \varepsilon_{t,i} \sim N(0, \sigma_t^2), \tag{5}$$

$$\delta_t(\mathbf{z}) = d_{t,0} + d_{t,1} z_c + d_{t,2} z_{f_{80}}, \quad d_{t,j} \sim N(0, \tau_t^2), \tag{6}$$

where $\mathbf{z}$ is the pair (setting, feed size) standardized with the mean and standard deviation of the training surveys, and $\tau_{\mathrm{tph}} = 140$ t/h and $\tau_{\mathrm{kW}} = 36$ kW (about 12 % of the typical level). Given $\boldsymbol{\theta}$ and σ_t the model is linear and Gaussian in $\mathbf{d}_t$, so the discrepancy is integrated out exactly: with $\mathbf{Z}$ the $n \times 3$ matrix of rows $(1, z_c, z_{f_{80}})$ of the training surveys,

$$\mathbf{y}_t \sim N\big(\mathbf{f}_t(\boldsymbol{\theta}),\ \sigma_t^2 \mathbf{I} + \tau_t^2 \mathbf{Z}\mathbf{Z}^\top\big)\,. \tag{7}$$

The two outputs are treated as independent given $\boldsymbol{\theta}$, and the likelihood is the product of the two densities in (7). The sampler explores the eight scalars and the two noise scales (ten dimensions). For each retained sample the discrepancy coefficients are drawn from their exact Gaussian conditional given the residual, and the posterior predictive of a held-out survey is the engine at that survey plus $\mathbf{x}_h^\top \mathbf{d}_t$ plus noise, with $\mathbf{x}_h$ the standardized regressors of the survey. The interval of 80 % is the 10th to the 90th percentile of 20 predictive draws per retained sample.

E. Sampler and convergence

The sampler is `emcee` 3.1.6 [14] with the differential-evolution move and the snooker move of ter Braak [15] and ter Braak and Vrugt [16], in proportions 0.8 and 0.2, with 20 walkers, 7000 steps, the first 1000 discarded and every 10th kept, which retains 12000 samples per posterior, and a fixed seed (42). The walkers start in a small ball around the prior centre (5 % of the prior standard deviation).

The affine-invariant stretch move of Goodman and Weare [13] was the first choice. On the same posterior it did not converge by the criterion below: the integrated autocorrelation time stayed above 100 steps and grew with the chain length, along the parameters that the surveys do not identify, so that no fold met the criterion at 7000 steps. The differential-evolution moves reach an autocorrelation time of about 50 steps on the same posterior within the same number of steps. The held-out scores hardly depend on the move: the power MAPE is 14.29 % with the stretch move and 14.31 % with the differential-evolution moves, and the throughput MAPE is 12.00 % with both.

A posterior counts as converged when the post-burn chain is at least 50 integrated autocorrelation times long, with the autocorrelation time of `emcee` (the largest over the ten dimensions). A posterior that fails the criterion is reported as not converged.

IV. EXPERIMENTS

A. Strict leave-one-survey-out

For each of the ten surveys, a posterior is computed from the other nine surveys alone: the engine scalars, the noise scales, the discrepancy and the standardization of the regressors are all refit inside the fold, and the held-out survey enters only the prediction. One more posterior is computed from all ten surveys for the identifiability table. That makes 11 posteriors.

B. Comparators

The three comparators are refit inside every fold on the nine training surveys.

- The *setting-only line* is an ordinary least-squares line of the output on the closed-side setting. It has no physics.
- The *refit backbone* is the engine at the nominal scalars with the capacity base Q_{ref} refit for throughput, and the slope and intercept of the form $P_c = \zeta P_d + P_n$ refit for power, both by least squares. Its 80 % band is the *residual band*: the prediction plus or minus 1.2816 times the standard deviation of the held-out residuals of the other nine surveys, so that a survey never enters the width of its own band.
- The *free-form linear fit* is an ordinary least-squares fit of the output on the setting and the feed size.

The bands of the setting-only line and of the free-form fit are built in the same way as the residual band. For power, Morrell's size-specific energy [8] is held out as well, with its work index refit per fold, with and without a no-load term; it reaches 38.08 % and 15.21 %, and its fitted work index is negative in every fold with the no-load term, so that variant is a descriptive line and not an estimate of the work index.

C. Scores

Per output: the mean absolute percentage error (MAPE) of the predictive mean; the root mean squared error (RMSE, in t/h or kW); the mean log predictive density of the held-out measurement (a log density per t/h or per kW, higher is better; for CrusherBayes the log of the mean of the Gaussian densities over the posterior draws, for the comparators a Gaussian with the residual-band standard deviation); the coverage of the 80 % interval, which is the fraction of surveys whose measurement lies inside; and the mean width of that interval. With $n = 10$ one survey moves a coverage by 0.1, and a difference of about 3.0 points of MAPE is within the noise.

D. Rule for dropping the method, stated before the run

Before the posteriors were computed we stated when the method would be dropped as a predictor: (a) its power MAPE is not below that of the setting-only line by at least 3.0

TABLE II
HELD-OUT SCORES OVER THE TEN SURVEYS (STRICT LEAVE-ONE-SURVEY-OUT). COVERAGE AND WIDTH ARE THOSE OF THE 80 % INTERVAL.

Target	Method	MAPE (%)	RMSE	Mean log density	Coverage	Width
Throughput (t/h)						
	CrusherBayes	12.00	168.57	−6.678	0.8	439.1
	Setting-only line	13.67	177.75	−6.709	0.8	478.8
	Refit backbone with residual band	12.09	176.67	−6.896	0.8	473.8
	Free-form linear fit	12.18	174.42	−6.647	0.8	468.5
Power (kW)						
	CrusherBayes	14.31	44.68	−5.346	0.9	117.7
	Setting-only line	12.78	44.56	−5.341	0.8	119.6
	Refit backbone with residual band	25.94	78.52	−5.847	0.9	209.8
	Free-form linear fit	14.03	48.68	−5.399	0.8	129.8

points and its throughput MAPE is not below that of the refit backbone by at least 3.0 points, and (b) on either output its coverage lies outside 0.7 to 0.9 or its mean band is wider than the residual band. Both conditions must hold. Whatever the outcome, the identifiability table is reported.

V. RESULTS

A. Held-out accuracy and calibration

Table II gives the scores. On throughput the posterior predictive mean has a MAPE of 12.00 %, against 12.09 % for the refit backbone, 12.18 % for the free-form fit and 13.67 % for the setting-only line; a constant mean gives 18.29 %. The differences of the first four are below the noise level of ten surveys. On power the posterior predictive mean has a MAPE of 14.31 %, against 25.94 % for the refit backbone, 14.03 % for the free-form fit and 12.78 % for the setting-only line, 12.78 % being the lowest; a constant mean gives 24.48 %. CrusherBayes therefore removes most of the error of the refit backbone on power, and it does not beat the line in the setting: its gain over the line is −1.53 points, which is within the noise. This is a measured negative result for accuracy.

The calibration is better than that of the backbone and comparable to that of the simple models. The 80 % interval of CrusherBayes covers 0.8 of the surveys for throughput and 0.9 for power, both inside 0.7 to 0.9. Its mean width is 439.12 t/h on throughput, 7.3 % narrower than the residual band of the refit backbone (473.76 t/h) at the same coverage, and 117.74 kW on power, 43.9 % narrower than the residual band (209.79 kW) at the same coverage of 0.9. On power the comparison with the simple models is different: the setting-only line has a mean width of 119.60 kW and covers 0.8 of the surveys, and the free-form fit has 129.82 kW and 0.8, against 117.74 kW and 0.9 for CrusherBayes. Its width is about that of the setting-only line, whose coverage is lower, so the gain in band width on power is against the backbone and not against the simple line. On throughput its width is smaller than that of the setting-only line (478.76 t/h) and of the free-form fit (468.51 t/h), all with coverage 0.8. Figure 1 shows the bands against the measured values survey by survey.

B. Outcome of the dropping rule

Table III applies the rule stated before the run. Condition (a) holds: the gain in power MAPE over the setting-only line is −1.53 points and the gain in throughput MAPE over the refit backbone is 0.09 points, both below 3.0. Condition (b) fails: the coverages 0.8 and 0.9 are inside the interval 0.7 to 0.9, and the bands are narrower than the residual band on both outputs. As both conditions are needed to drop the method, the rule does not drop it. The rule, as written, therefore retains a method that calibrates its band and does not improve the point prediction; it does not assert a gain in accuracy.

TABLE III
THE TERMS OF THE RULE FOR DROPPING THE METHOD AS A PREDICTOR, EVALUATED ON THE HELD-OUT SCORES. THE REFERENCE IS THE SETTING-ONLY LINE FOR POWER AND THE REFIT BACKBONE FOR THROUGHPUT.

Term	Throughput	Power
MAPE of CrusherBayes (%)	12.00	14.31
MAPE of the reference (%)	12.09	12.78
Gain over the reference (points)	0.09	−1.53
Gain of at least 3 points	no	no
Coverage	0.8	0.9
Coverage outside 0.7 to 0.9	no	no
Mean width (t/h, kW)	439.1	117.7
Mean width of the residual band	473.8	209.8
Band wider than the residual band	no	no

C. What the surveys identify

Table IV compares the prior with the posterior of every parameter, fitted to all ten surveys. The ratio is the posterior standard deviation over the prior standard deviation; a ratio far below one means that the surveys determine the parameter, a ratio near one that it stays at its prior. We read a ratio up to 0.5 as identified, up to 0.9 as partly identified and above 0.9 as at its prior; a ratio slightly above one (1.026 for s_{K_2}) is read as no contraction. Figure 2 shows the prior and a normal summary of the posterior for each parameter, and Fig. 3 the ratios.

Of the 16 parameters, 7 are identified, 4 are partly identified and 5 stay at their prior. The identified ones are the capacity base Q_{ref} (ratio 0.454), the two noise scales (0.301 for throughput, 0.332 for power) and the four discrepancy slopes on the setting and the feed size (0.350 to 0.381). Partly identified are ζ (0.733), P_n (0.795), W_i (0.854) and the constant of the throughput discrepancy (0.878). Four of the eight engine scalars stay at their prior: the multipliers s_{K_1} (0.965) and s_{K_2} (1.026), the exponent K_3 (0.992) and the breakage shift Δt_{10} (0.990), and so does the constant of the power discrepancy (0.910). The posterior mean of Q_{ref} is 646.35 t/h against a nominal 647 t/h: the surveys narrow the capacity base and leave its centre where it was.

D. The discrepancy

The discrepancy slope on the setting is large for both outputs. For power it is -57.07 ± 13.70 kW per standard deviation of the setting (about −9.6 kW/mm, with 5.9 mm the standard deviation of the setting over the ten surveys), and for throughput 100.84 ± 49.00 t/h per standard deviation (about 17.0 t/h per mm). The measured power falls as the setting widens (correlation −0.82), whereas the calibrated engine power rises with it (correlation 0.79, from 365 to 419 kW

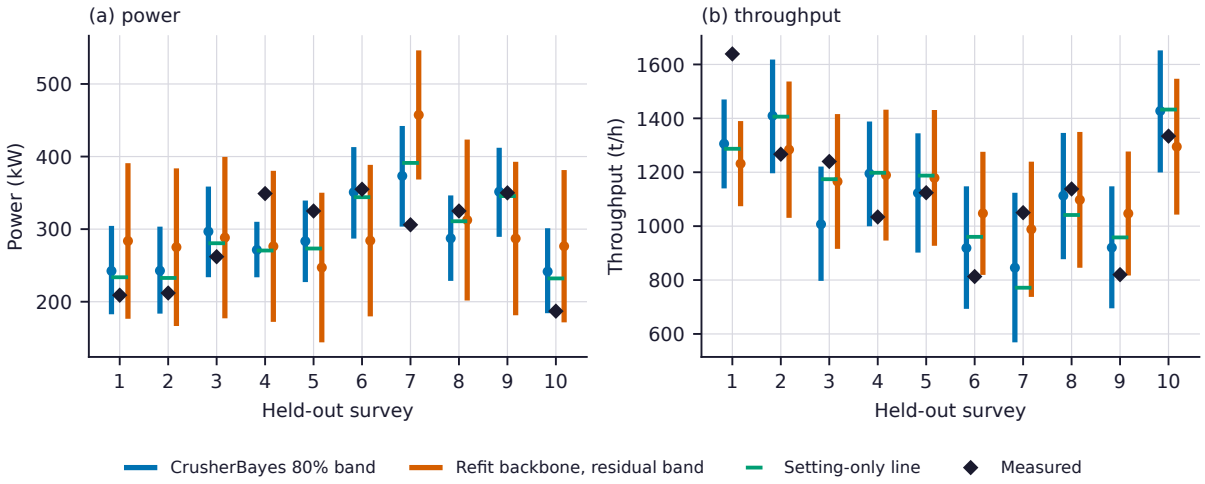


Fig. 1. Held-out 80 % predictive interval against the measured value, per survey, for (a) power and (b) throughput. The blue interval is the posterior predictive of CrusherBayes and the orange interval the residual band of the refit backbone; the green dash is the setting-only line and the black diamond the measurement. Every interval is computed without the survey it is drawn for.

TABLE IV
PRIOR AGAINST POSTERIOR OF THE 16 PARAMETERS, FITTED TO ALL TEN SURVEYS. THE RATIO IS THE POSTERIOR STANDARD DEVIATION OVER THE PRIOR STANDARD DEVIATION (NATURAL SCALE).

Parameter	Unit	Prior mean	Prior SD	Posterior mean	Posterior SD	Ratio	Reading
Q_{ref}	t/h	647.00	167.01	646.35	75.74	0.454	identified
W_i	kWh/t	9.00	1.52	8.39	1.29	0.854	partly identified
ζ	–	1.30	0.30	1.14	0.22	0.733	partly identified
P_n	kW	110.00	47.85	86.35	38.04	0.795	partly identified
s_{K_1}	–	1.00	0.15	1.07	0.15	0.965	at its prior
s_{K_2}	–	1.00	0.15	1.05	0.16	1.026	at its prior
K_3	–	2.30	0.49	2.23	0.49	0.992	at its prior
Δt_{10}	points	0.00	3.99	−0.11	3.95	0.990	at its prior
σ_{tph}	t/h	60.00	129.67	145.23	39.04	0.301	identified
σ_{kW}	kW	15.00	32.42	38.81	10.75	0.332	identified
$d_{\mathrm{tph},0}$	t/h	0.00	140.00	53.36	122.88	0.878	partly identified
$d_{\mathrm{tph},1}$ (setting)	t/h	0.00	140.00	100.84	49.00	0.350	identified
$d_{\mathrm{tph},2}$ (f_{80})	t/h	0.00	140.00	56.57	47.63	0.340	identified
$d_{\mathrm{kW},0}$	kW	0.00	36.00	−12.57	32.77	0.910	at its prior
$d_{\mathrm{kW},1}$ (setting)	kW	0.00	36.00	−57.07	13.70	0.381	identified
$d_{\mathrm{kW},2}$ (f_{80})	kW	0.00	36.00	−5.16	13.03	0.362	identified

over the ten surveys); the negative discrepancy slope of power has the opposite sign to the setting dependence of the engine, and compensates it.

E. Convergence

All 11 posteriors meet the criterion (Table VI of Appendix C): the largest autocorrelation time is between 48.0 and 84.8 steps and the 6000 post-burn steps are 70.7 to 125.0 times it. The acceptance fraction is between 0.171 and 0.204.

VI. DISCUSSION

a) Why the point accuracy does not improve: Throughput of the engine is $Q_{\mathrm{ref}}\sigma(c)$ with σ linear in the setting, and the discrepancy adds a term linear in the setting and the feed size. Measured power correlates with the setting at −0.82, and the engine's own power moves the other way. Once the discrepancy has absorbed that setting dependence, the posterior predictive is close in form to a line in the setting and the feed size, which the setting-only line and the free-form fit already are; this agrees with the ties in Table II, but we did not test it directly.

b) Why the population balance is not identified: The two measured outputs carry the product size only through the Bond term of power, and power follows the setting and the throughput, not the product size. The four scalars that set the product size (s_{K_1}, s_{K_2}, K_3, Δt_{10}) therefore remain at their prior, and the engine's product size, which is not measured in the ten surveys, is not validated by them. Measured product size distributions per survey would constrain these scalars directly. We did not test that, because the distributions of the source are figures only.

c) Confounding of calibration and discrepancy: Brynjarsdóttir and O'Hagan [12] show that the discrepancy is confounded with the calibration parameters. Two instances appear here. The capacity base and the constant of the throughput

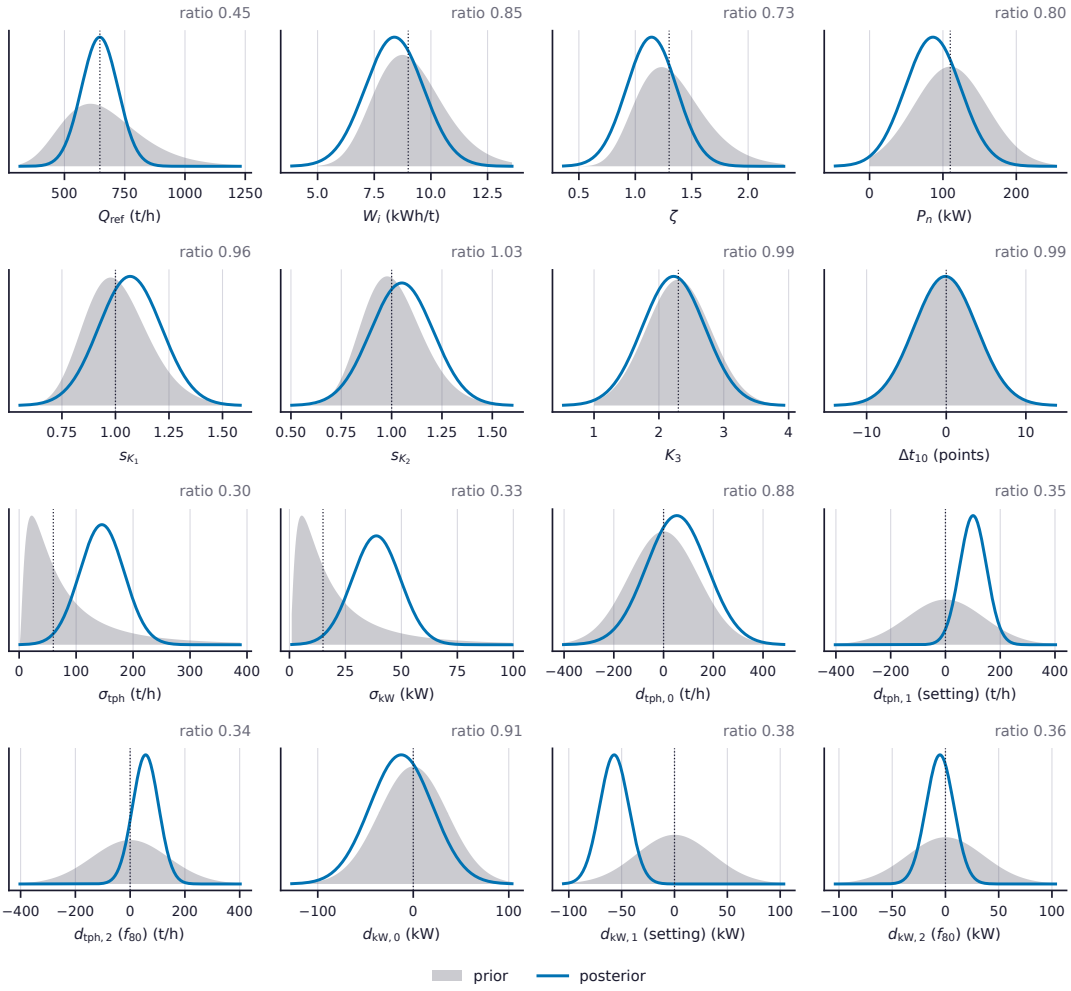


Fig. 2. Prior (grey area) against posterior (blue line) of each parameter, fitted to all ten surveys. The posterior is drawn as a normal density with its mean and standard deviation, the two summaries of the fit; the dotted line is the nominal value. The ratio is the posterior over the prior standard deviation.

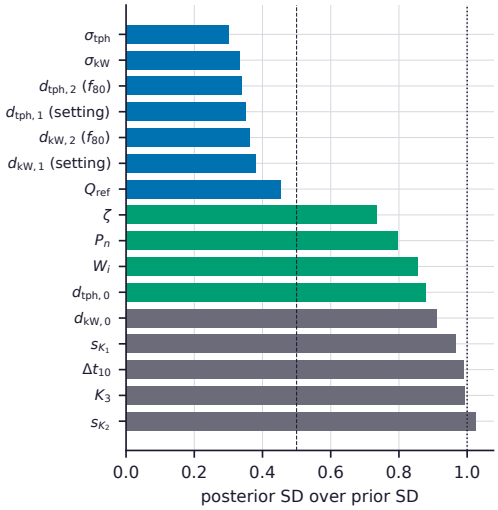


Fig. 3. Posterior over prior standard deviation of each parameter: identified (blue, up to 0.5), partly identified (green, up to 0.9) and at the prior (grey). The dashed line marks 0.5 and the dotted line 1.

discrepancy both set the level of throughput, and only the first contracts strongly (0.454 against 0.878). And since W_i and ζ multiply, power fixes their product and not each of them, which is why both are only partly identified. The posterior of the individual scalars should not be read as an estimate of a physical value of the ore.

d) What the band is worth: The band of the setting-only line, built from the spread of its held-out residuals, is as wide as the CrusherBayes band on power (119.60 against 117.74 kW), with a coverage of 0.8 against 0.9. The advantage of CrusherBayes over the line is that its band comes from a generative model of the survey (engine, discrepancy and noise), so the band carries the uncertainty of the scalars; whether it stays calibrated at a setting outside the surveyed range was not tested. The advantage over the refit backbone is plain in the data: the same coverage with a band 43.9 % narrower on power. Against the setting-only line the power width is about the same (117.74 against 119.60 kW) and the coverage is higher (0.9 against 0.8), a difference of one survey out of ten; the band gain is not against the simple line.

VII. Limitations

- Ten surveys of one machine, one ore and one operation, with two measured outputs and two regressors. Coverage is coarse: one survey moves it by 0.1, and the coverages of 0.8 and 0.9 do not distinguish a calibrated band from a slightly miscalibrated one.
- The widths of the priors and the discrepancy standard deviations τ_t are our choices; no sensitivity analysis of the held-out scores or of the ratios to those choices was made, and the confounding of discrepancy and parameters is resolved only by meaningful priors [12].
- The discrepancy is linear in two standardized regressors with a fixed prior scale, not a Gaussian process with estimated hyperparameters; the two outputs are independent given the scalars.
- The throw and the speed are fixed at the machine's nominal values, and the feed gradation is reconstructed from the measured f_{80} with a Rosin–Rammler distribution of fixed uniformity, so neither the throw, the speed nor the feed shape is calibrated.
- The convergence criterion uses the autocorrelation time of a single ensemble with one seed, and we did not run independent ensembles with other seeds. The comparison with the stretch move is a measurement on the same posterior, reported from the preliminary runs, and was not repeated for this paper.
- The rule for dropping the method is lenient: it drops a method only when it fails on accuracy and on calibration at once, so it retains a method that does not improve accuracy.
- The comparison with other crusher models is limited to the comparators of the Experiments section; the discrete-element simulation of a crusher is not covered, and the search of the literature was not exhaustive.

VIII. Conclusions

Ten HP500 surveys, two measured outputs and a linear discrepancy identify the capacity base, the two noise scales and the dependence of both outputs on the setting and the feed size. They determine the work index and the power form only as a product, and they leave the classification and breakage scalars of the population balance at their prior. Strict leave-one-survey-out validation shows a throughput MAPE of 12.00 % and a power MAPE of 14.31 % for the posterior predictive mean, no better than a line in the closed-side setting (13.67 % and 12.78 %), with 80 % bands that are 43.9 % narrower on power than the residual band of the refit backbone at the same coverage; against the setting-only line the power band has about the same width (117.74 against 119.60 kW) and a coverage of 0.9 against 0.8. The finding is negative for accuracy and positive for calibration and for identifiability: to calibrate the population balance itself, the surveys need to measure the product size.

Appendix A
Reproducibility

The artifacts of the paper are in the repository `github.com/fsantibanezleal/CAOS_ChancaDEM`. The posteriors are sampled offline by the pipeline step of the data bake (`python data-pipeline/run.py hp500`), which writes `data/derived/hp500/crusherbayes.json`; the engine, the priors, the likelihood, the sampler and the scores are in `data-pipeline/pipeline/hp500/crusherbayes.py`. Every figure, table and number of this paper is regenerated from the committed artifacts `crusherbayes.json`, `loo.json`, `backbone.json` and `data/examples/hp500-surveys.csv` by `python manuscripts/crusherbayes/figures/make_figs.py`, which also writes the number-to-key table `manuscripts/crusherbayes/data/claims.json`. The sampler is `emcee` 3.1.6 with seed 42. The document carries the SHA-256 of its inputs,

```
483d24c5ddd35b41f2be05628fe9f09fd22562c5ab59cda69e036b9bb0ce605a
```

and the SHA-256 of the file read for this paper is

```
2ad1ae736c7c8a6559be448ea6fa67b732d051e17ea7d8be7487863f7e0fb96c
```

Appendix B
Survey data

TABLE V
THE TEN HP500 SURVEYS (ROCHA ET AL. [6], TABLE A1, CC BY 4.0): CLOSED-SIDE SETTING, FEED f_{80}, THROUGHPUT AND MOTOR POWER.

Survey	Setting (mm)	f_{80} (mm)	Throughput (t/h)	Power (kW)
1	54	88	1639	209
2	54	84	1267	212
3	48	47	1240	262
4	48	88	1034	349
5	48	62	1124	325
6	40	79	813	355
7	38	140	1050	306
8	44	123	1138	325
9	40	81	820	350
10	55	81	1334	187

Appendix C
Convergence per posterior

TABLE VI
CONVERGENCE OF THE 11 POSTERIORS: THE LARGEST INTEGRATED AUTOCORRELATION TIME $\tau_{\max}$ OVER THE TEN DIMENSIONS, THE POST-BURN CHAIN LENGTH IN UNITS OF $\tau_{\max}$ (THE CRITERION IS 50), THE MEAN ACCEPTANCE FRACTION AND THE RETAINED SAMPLES.

Posterior	$\tau_{\max}$ (steps)	Post-burn steps / $\tau_{\max}$	Acceptance	Retained samples
without survey 1	52.6	114.1	0.204	12000
without survey 2	52.7	113.8	0.201	12000
without survey 3	54.7	109.7	0.203	12000
without survey 4	84.8	70.7	0.171	12000
without survey 5	52.7	113.9	0.203	12000
without survey 6	48.0	125.0	0.204	12000
without survey 7	55.1	108.9	0.200	12000
without survey 8	48.9	122.7	0.204	12000
without survey 9	52.6	114.1	0.203	12000
without survey 10	50.7	118.3	0.202	12000
all ten surveys	48.6	123.5	0.204	12000

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